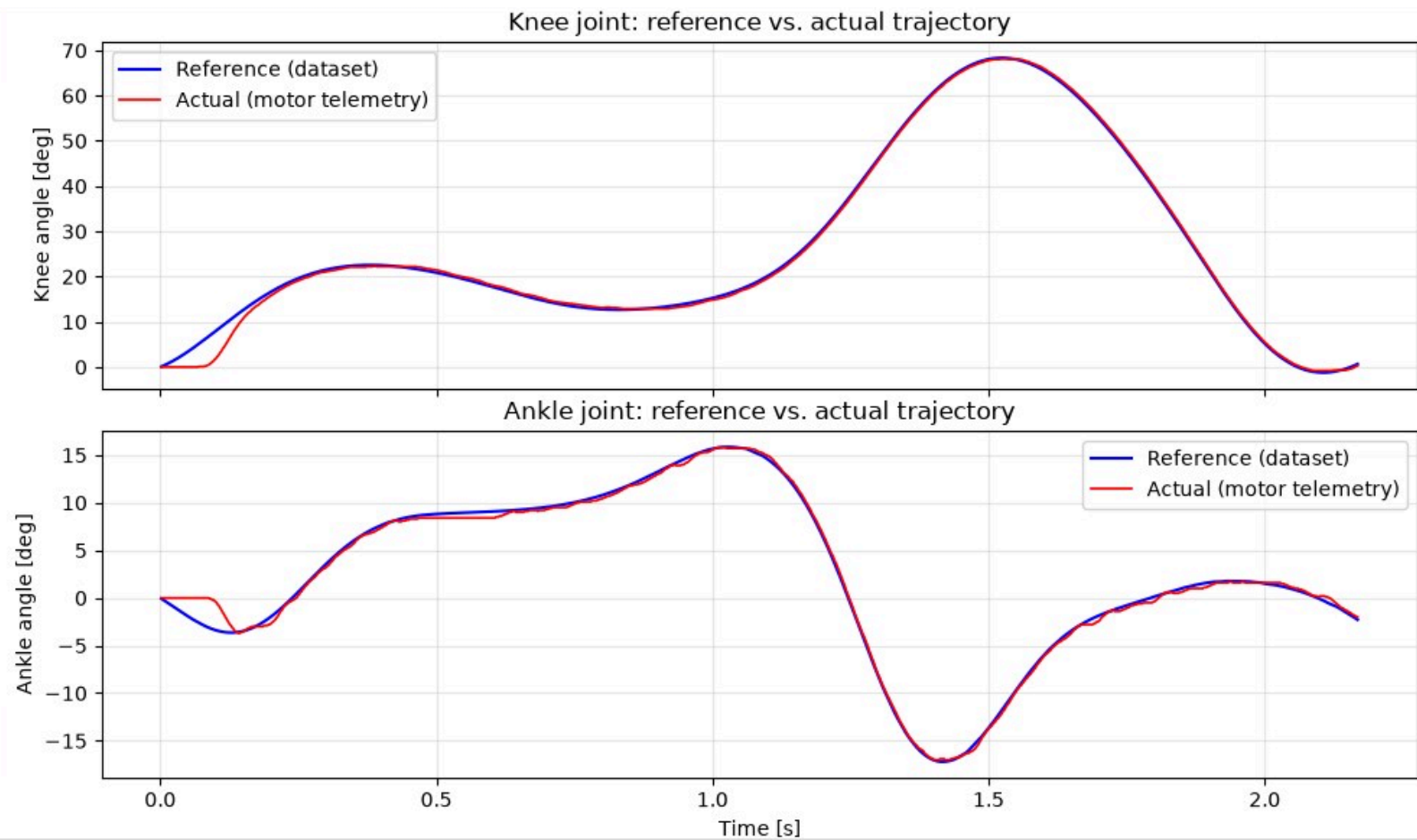
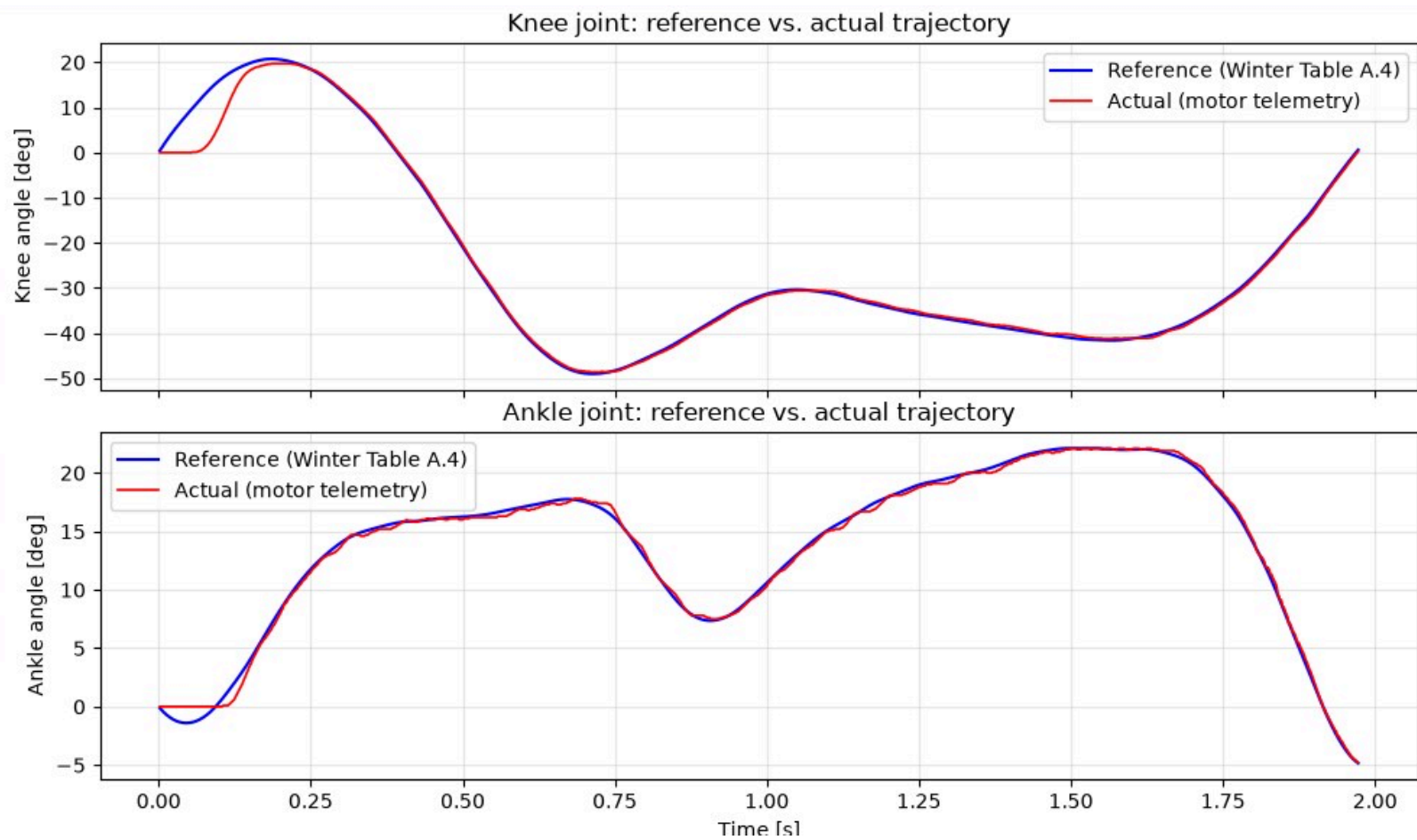


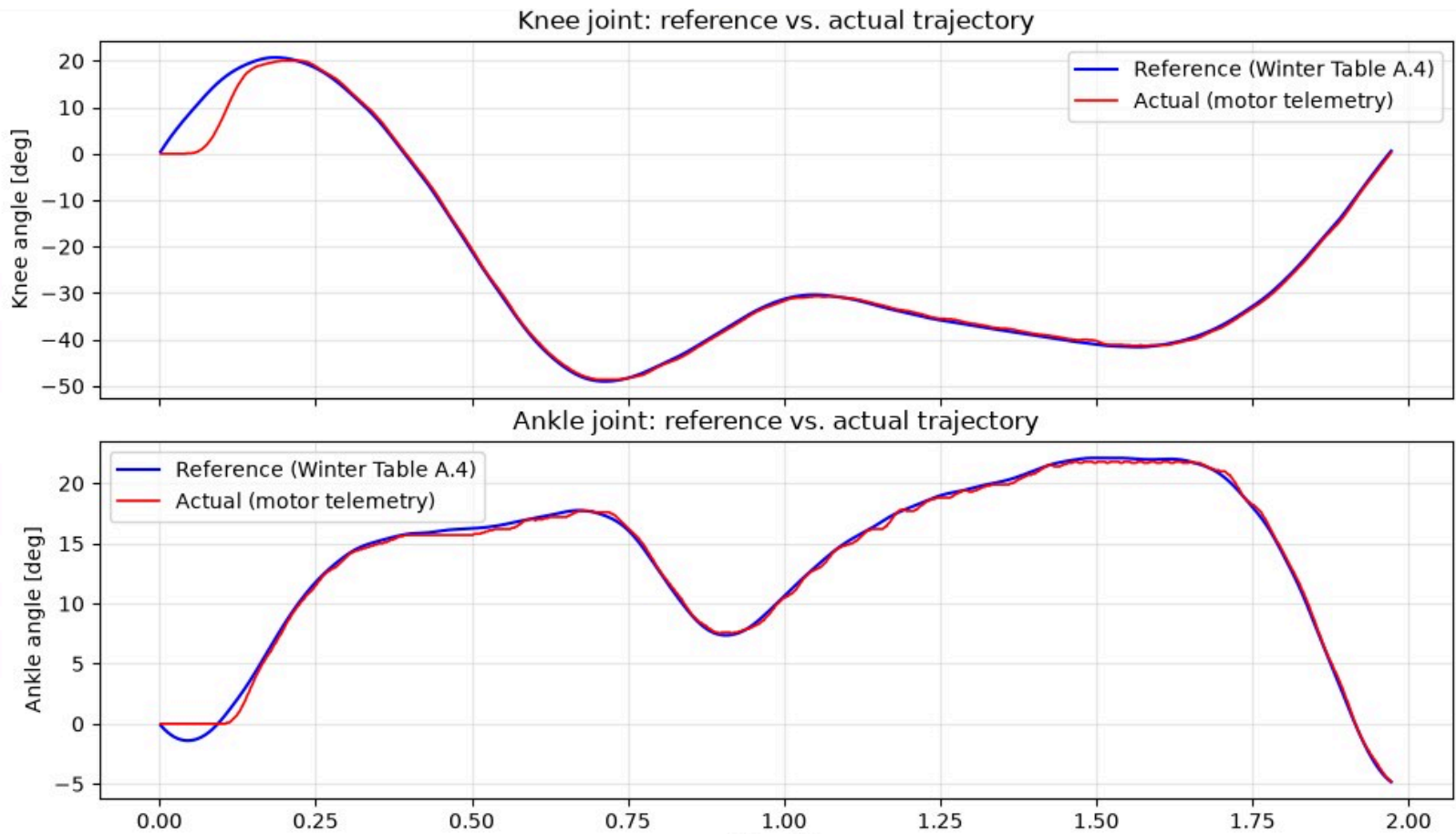
Knee-Ankle Trajectory Following Using Impedance Control Without Velocity Filter :
No-Load Condition (Knee Kp = 21.0, Knee Kd = 0.8 , Ankle Kp = 40.0, Ankle Kd = 0.8)
(Best Results)



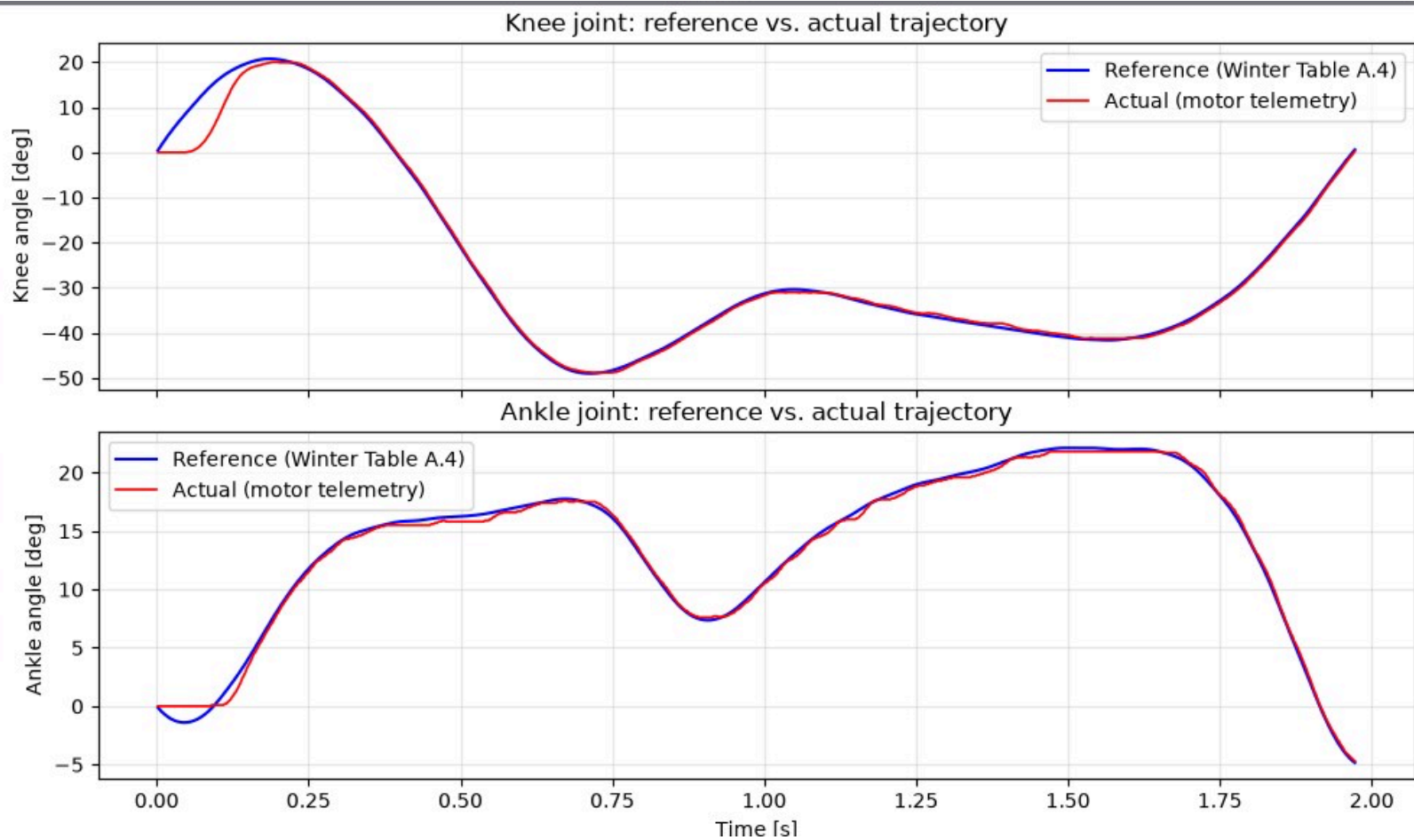
Knee-Ankle Trajectory Following Using Impedance Control With [Butterworth Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 224 Hz)



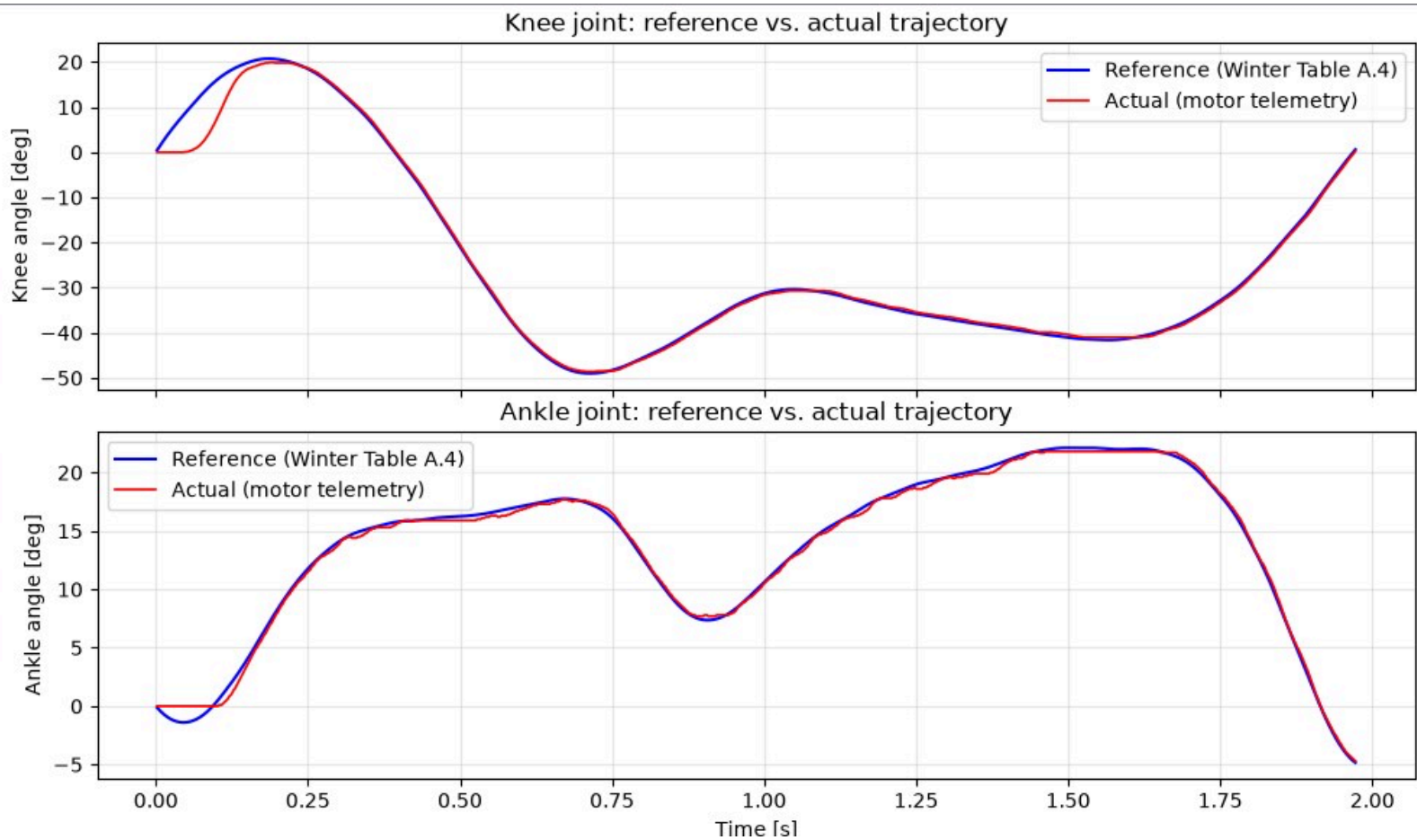
Knee-Ankle Trajectory Following Using Impedance Control With [Butterworth Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 210 Hz)



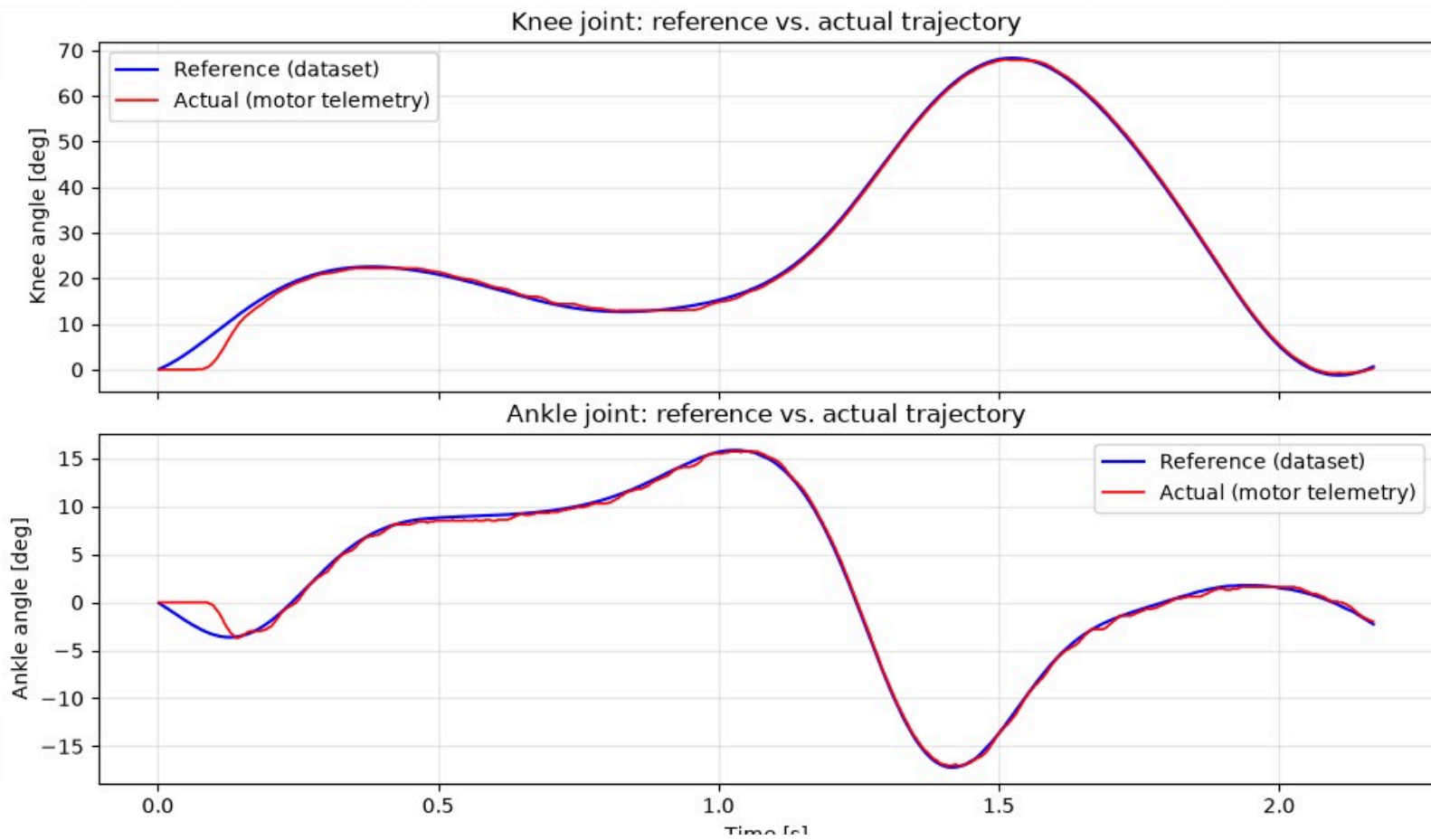
Knee-Ankle Trajectory Following Using Impedance Control With [Butterworth Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 200 Hz)



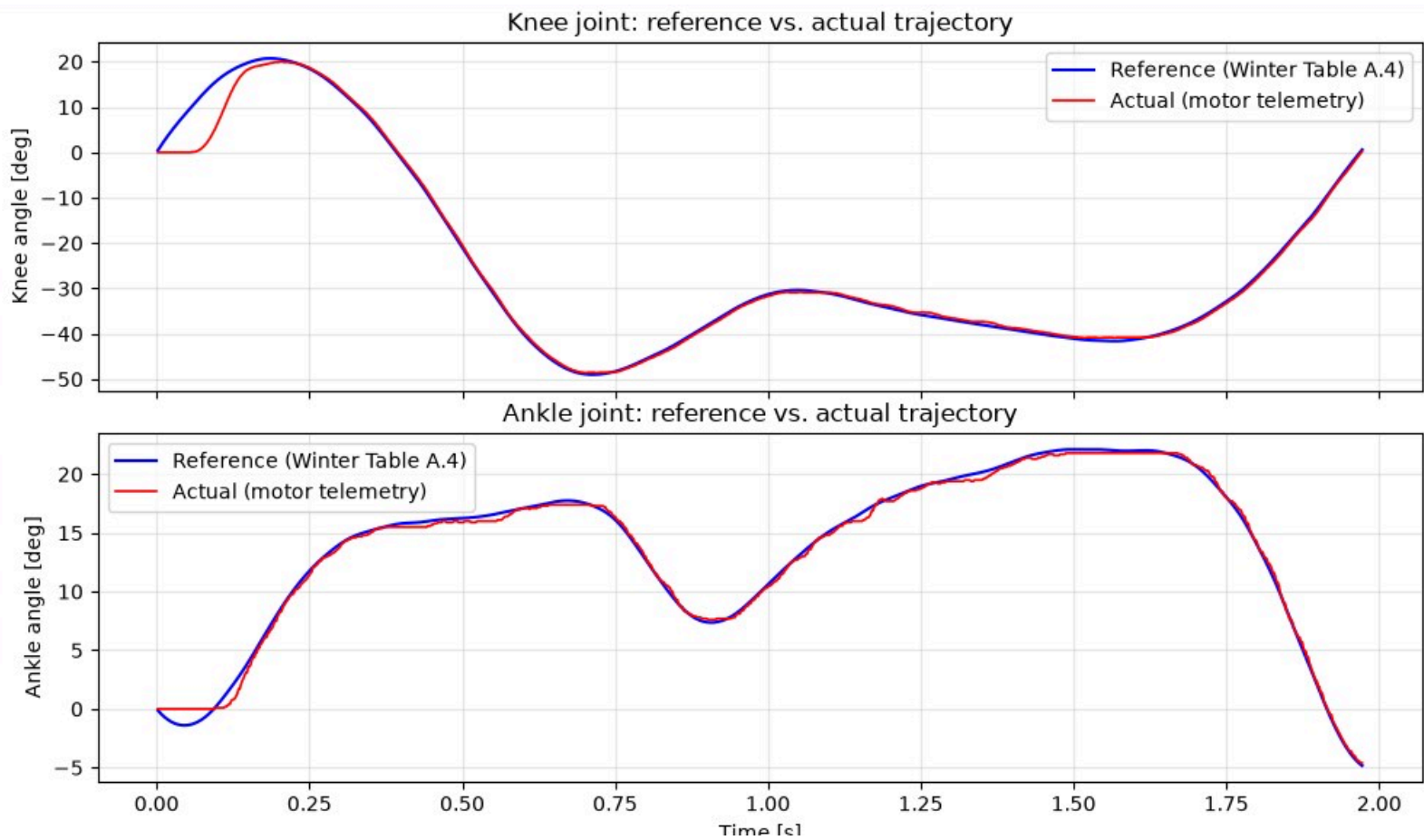
Knee-Ankle Trajectory Following Using Impedance Control With [Butterworth Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 150 Hz)



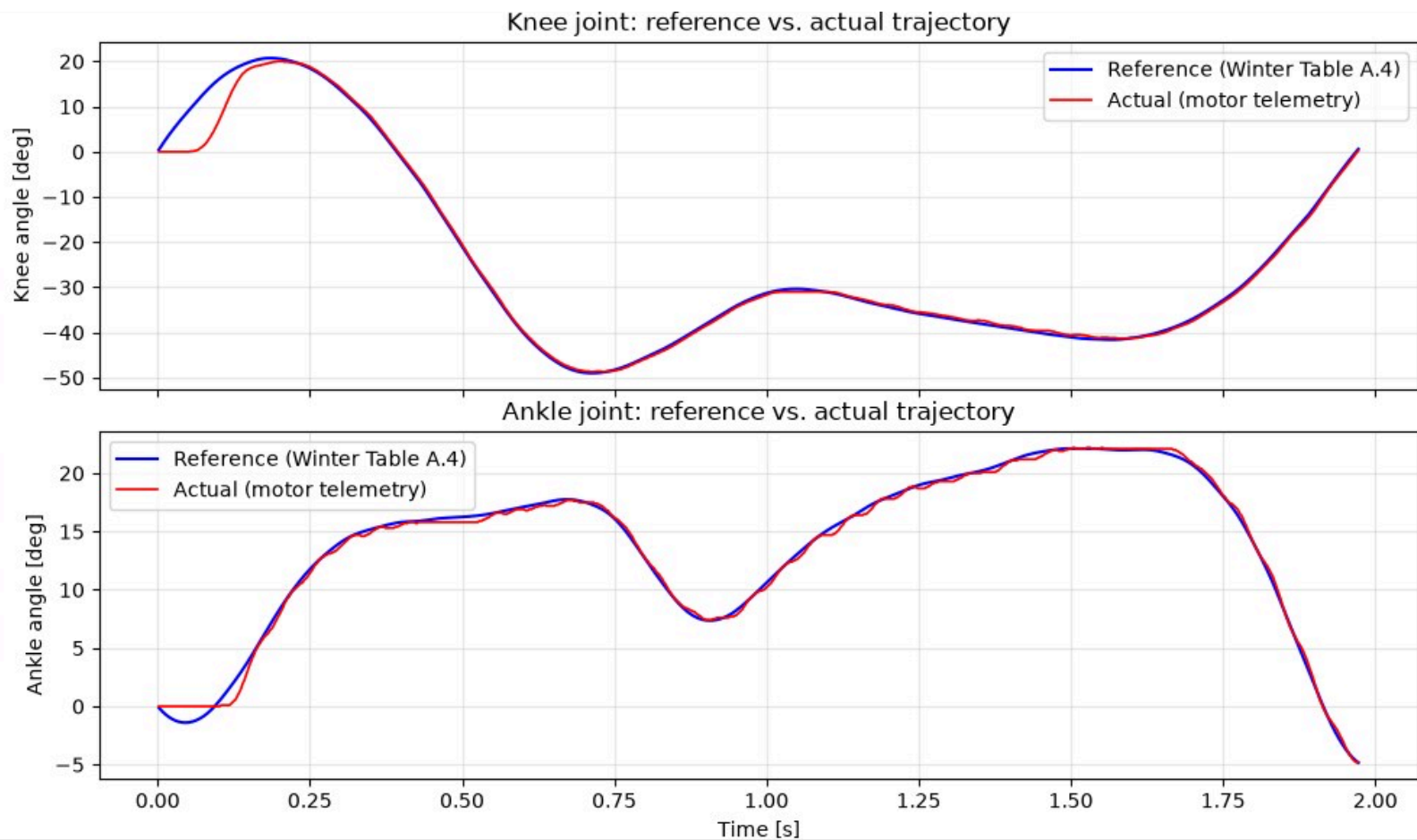
Knee-Ankle Trajectory Following Using Impedance Control With [Butterworth Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 140 Hz) (Best Results)



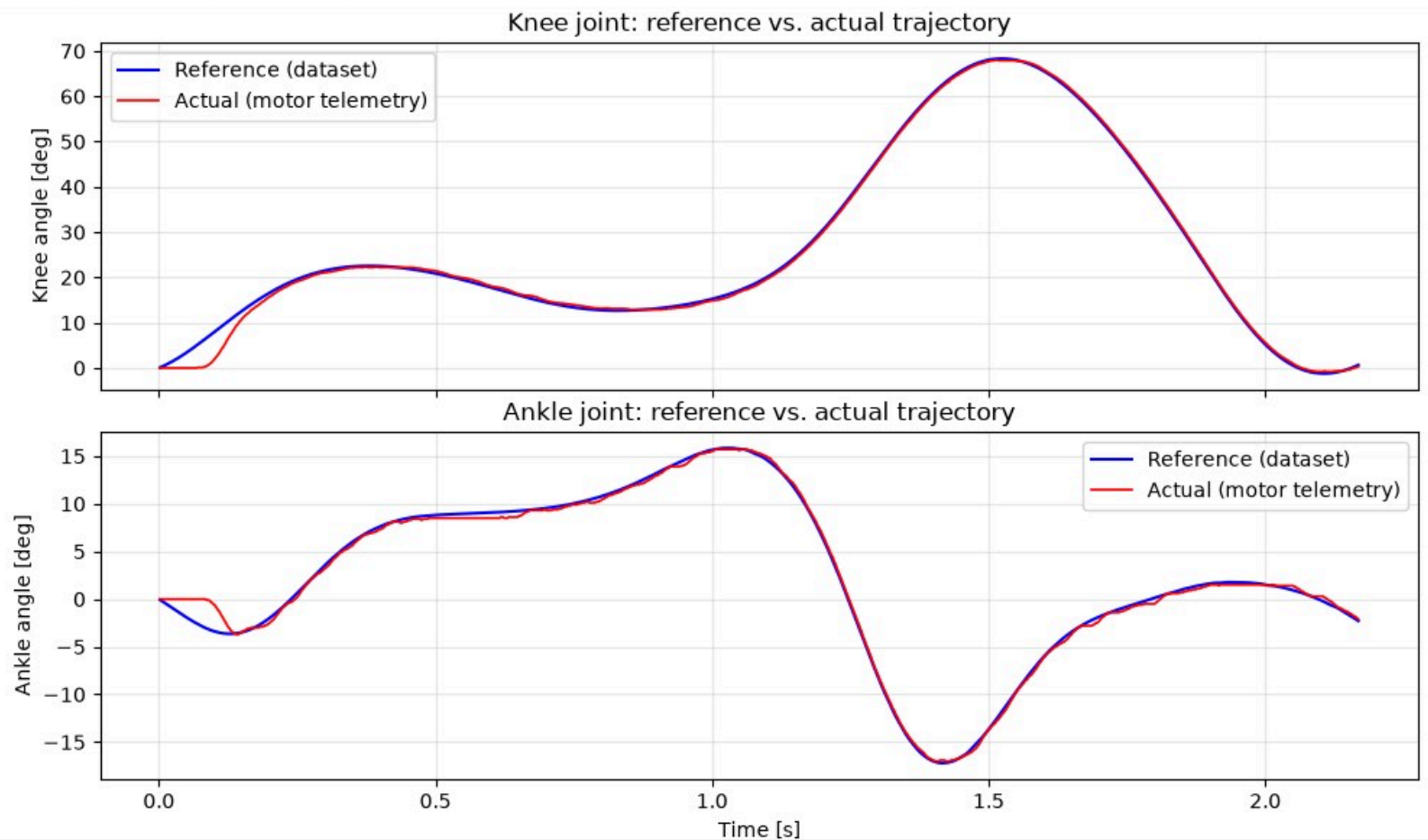
Knee-Ankle Trajectory Following Using Impedance Control With [Butterworth Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 100 Hz)



Knee-Ankle Trajectory Following Using Impedance Control With [Butterworth Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 80 Hz)

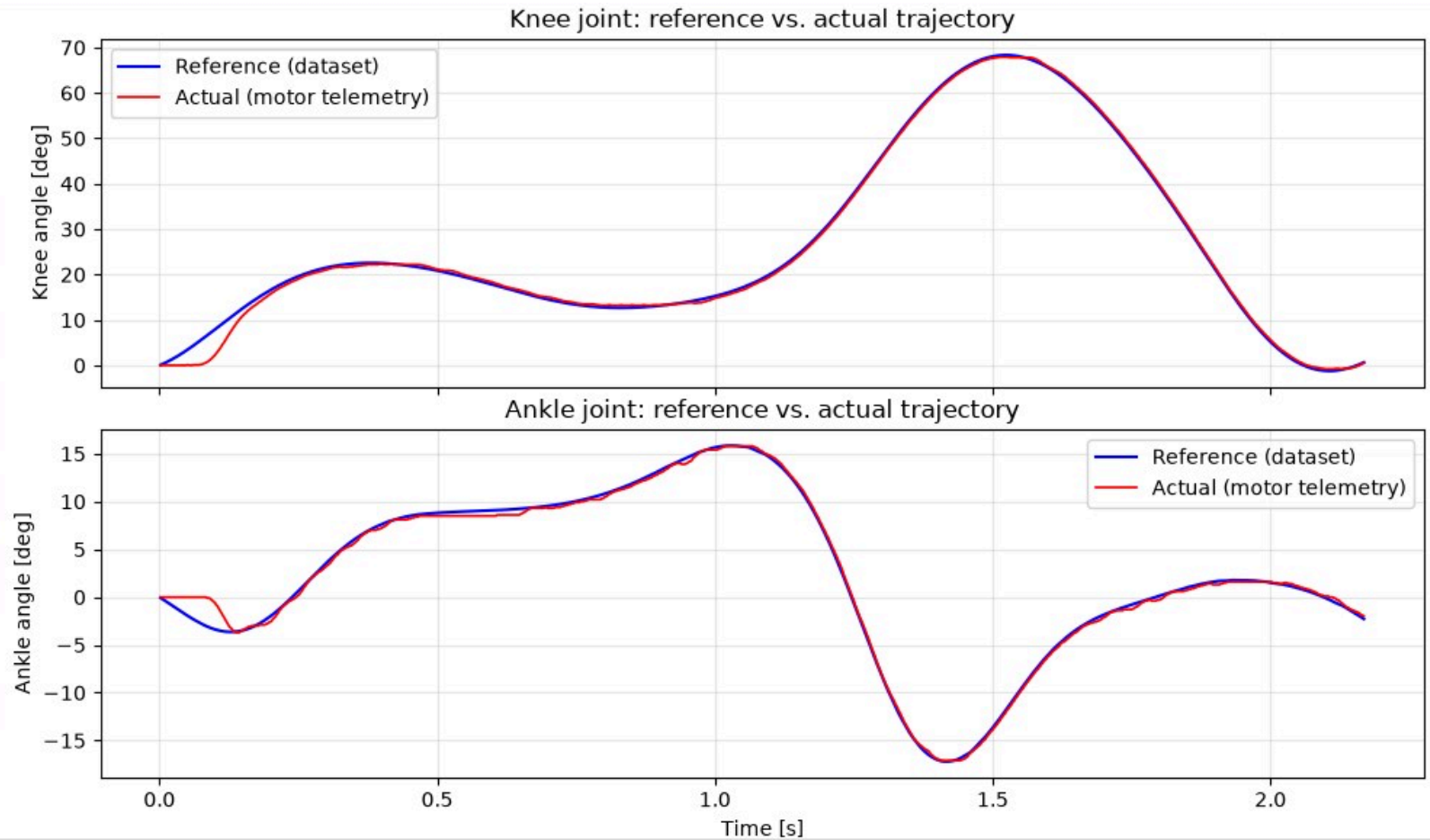


Knee-Ankle Trajectory Following Using Impedance Control With [Exponential Moving Average Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 140 Hz)



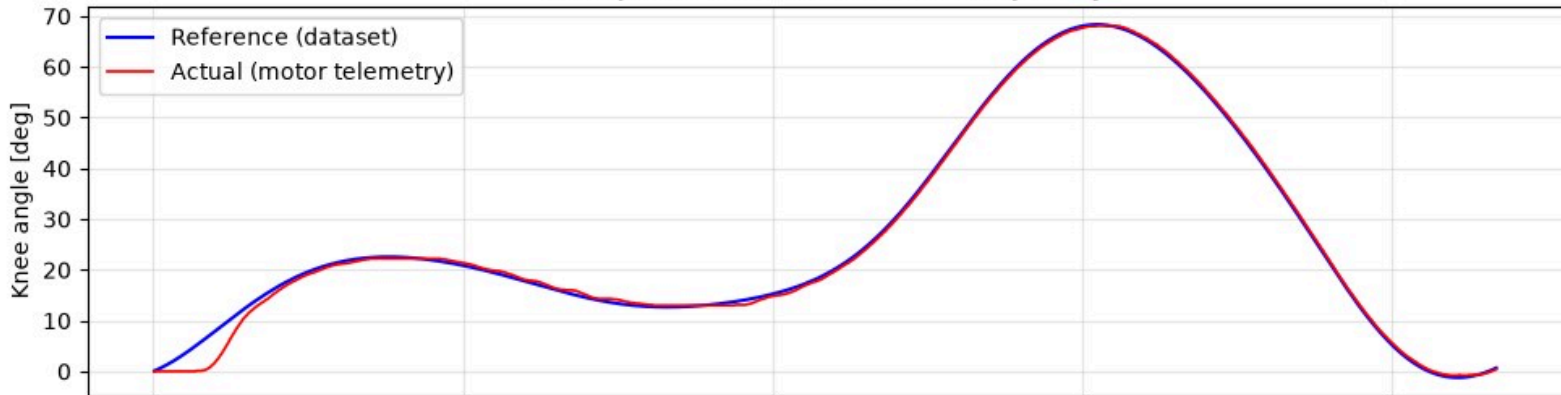
(x, y) = (1.218, -9.0)

Knee-Ankle Trajectory Following Using Impedance Control With [Exponential Moving Average Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 120 Hz) (Best Results)

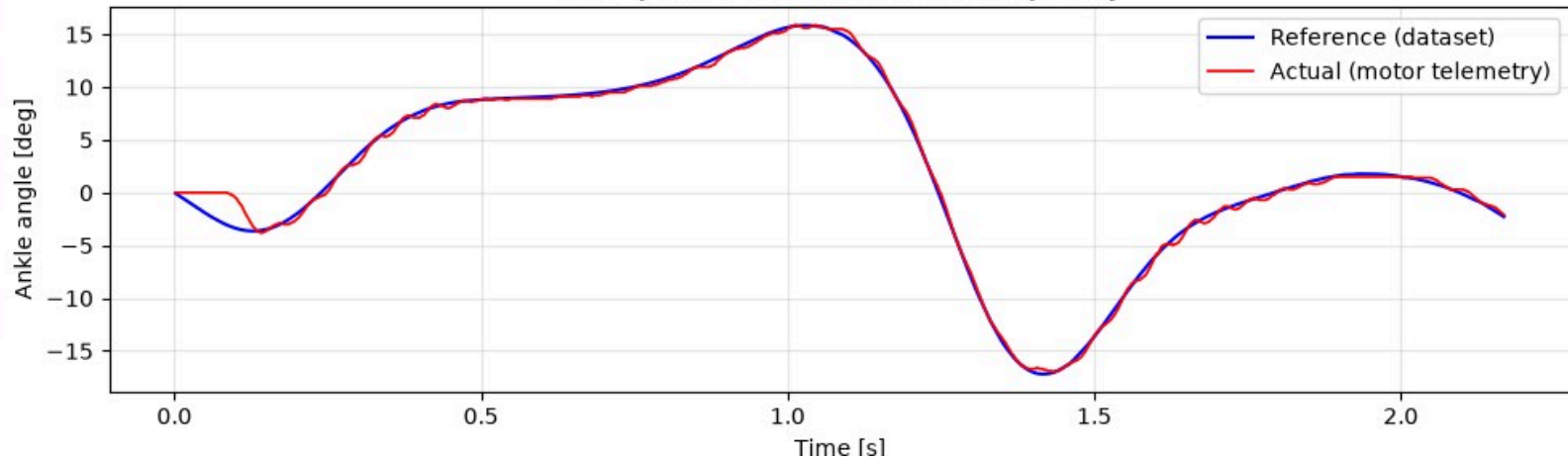


Knee-Ankle Trajectory Following Using Impedance Control With [Exponential Moving Average Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 100 Hz)

Knee joint: reference vs. actual trajectory

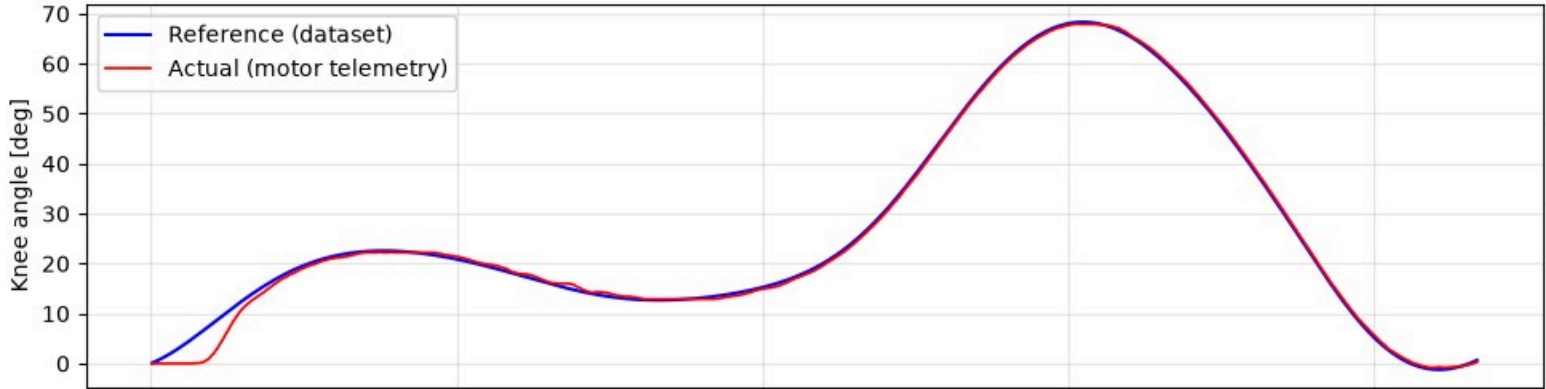


Ankle joint: reference vs. actual trajectory

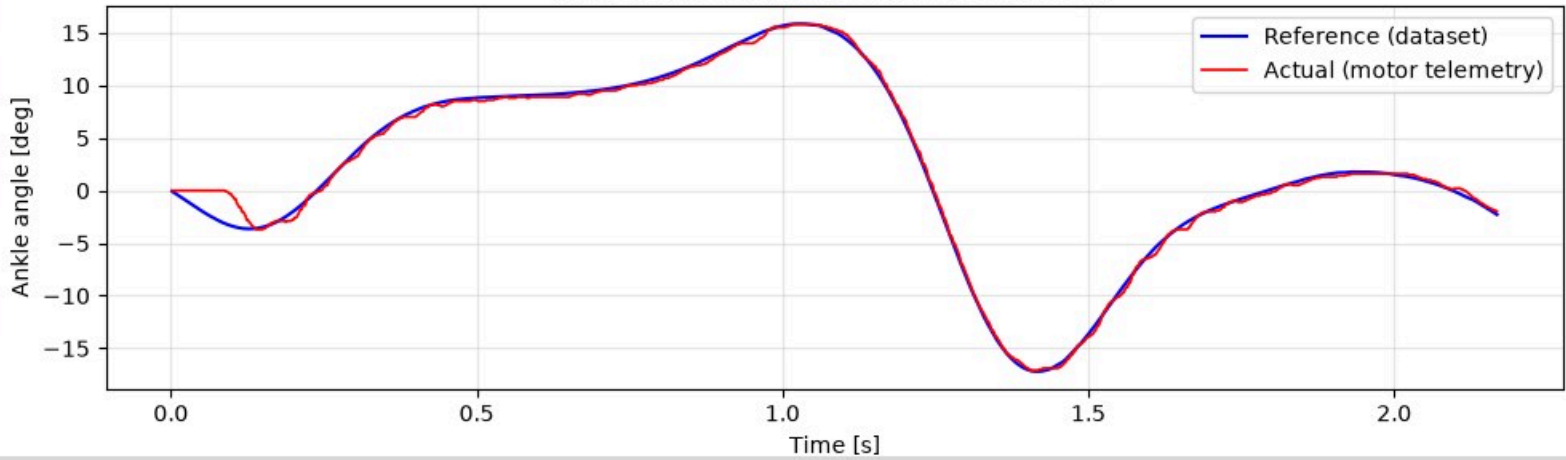


Knee-Ankle Trajectory Following Using Impedance Control With [Exponential Moving Average](#) Low-Pass Velocity Filter : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 80 Hz)

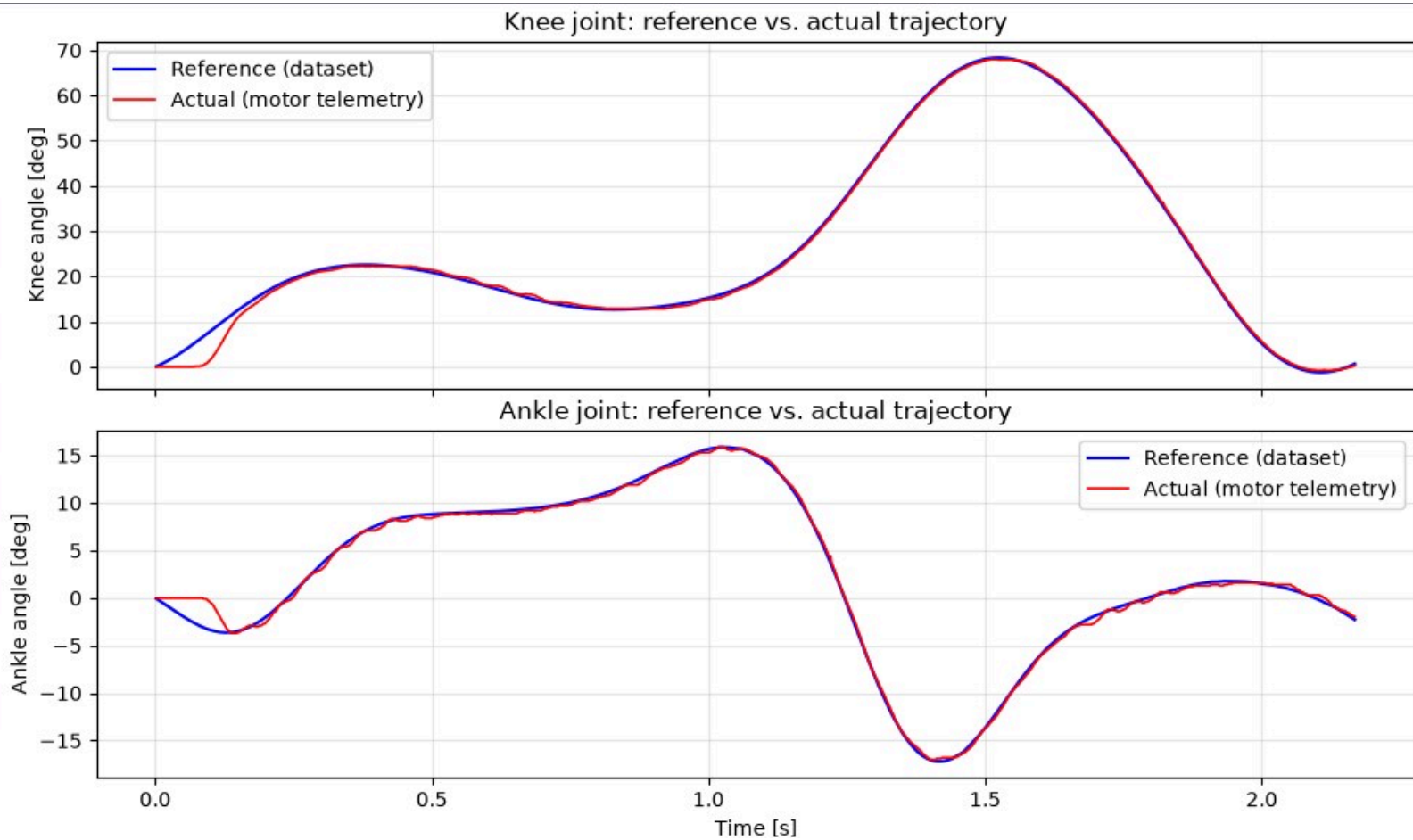
Knee joint: reference vs. actual trajectory



Ankle joint: reference vs. actual trajectory

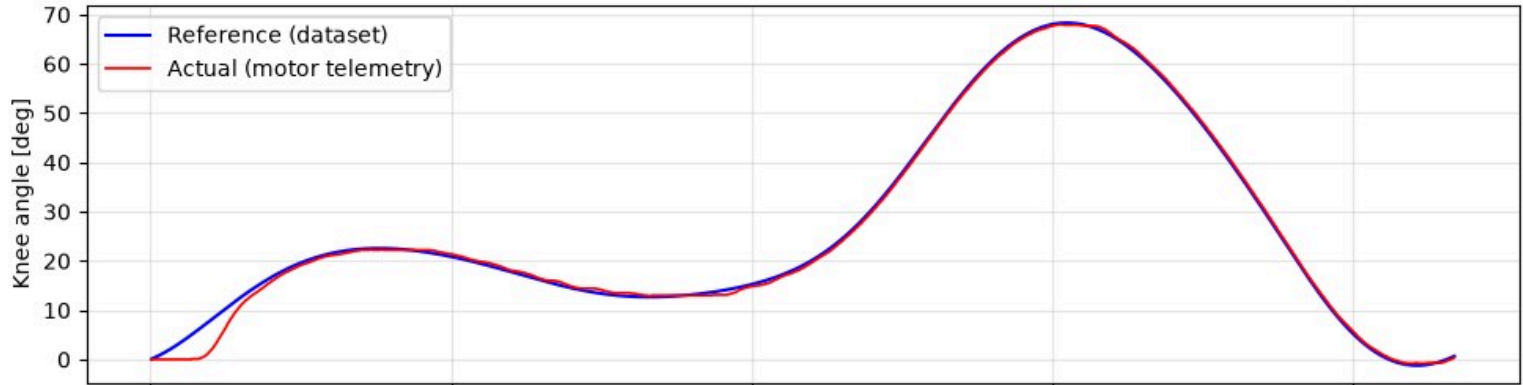


Knee-Ankle Trajectory Following Using Impedance Control With [Exponential Moving Average Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 60 Hz)



Knee-Ankle Trajectory Following Using Impedance Control With [Exponential Moving Average Low-Pass Velocity Filter](#) : In No-Load Condition (Controller Kp-Kd values same as above; Filter Order = 1, Knee Cutoff = 100Hz, Ankle Cutoff = 40 Hz)

Knee joint: reference vs. actual trajectory



Ankle joint: reference vs. actual trajectory

